Unified Salt–Ice–Water Model

Full LaTeX Equation Compendium and Derivations

Dynamic evolution, static equilibrium, Styrofoam heat transfer, ice melting and NaCl dissolution

SYPT modelling notes

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Abstract

This document expands the equations developed throughout our conversation into a unified model. It preserves the original successive formulations, derives the fully coupled time-dependent system, distinguishes insulated equilibrium from room-exposed long-time equilibrium, and explains the approximation limits. Equations from earlier stages that need qualification or correction are explicitly labelled; they should *not* be used indiscriminately as exact thermodynamic identities. No Mathematica implementation is included.

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1 System, notation and assumptions

A closed, rigid container contains ice, liquid water (or brine), solid NaCl and dissolved NaCl. Let $m_i(t)$, $m_w(t)$ and $m_s(t)$ denote the masses of ice, liquid water and *dissolved* NaCl respectively (kg). Let $m_{s,0}$ denote total initially added solid salt (kg), with $m_{s,d,0}$ denoting any initially dissolved salt. Define $M_w = m_{i,0} + m_{w,0}$ as total water mass, and $S_0 = m_{s,0} + m_{s,d,0}$ as total salt mass. All temperatures T in sensible-heat differences may be reported in Celsius (increments are kelvins); absolute thermodynamic energy references are specified later.

Symbol	Definition	SI units
k	Styrofoam conductivity	W/(m K)
A	Effective conduction area	m^2
d	Thickness of insulating layer	m
$H = kA/d$	Effective conductance	W/K
T, T_{room}	Mixture and room temperatures	°C
c_i, c_w	Ice and liquid-water specific heat capacities	J/(kg K)
L_f	Latent heat of melting water	J/kg
M_{NaCl}	NaCl molar mass	kg/mol
ΔH_{sol}	Molar enthalpy of NaCl dissolution	J/mol
K_f	Cryoscopic constant of water	K kg/mol
i_v	van 't Hoff factor (renamed i_v to avoid confusion with ice index)	dimensionless
b	Salt molality, mol per kg of liquid water	mol/kg
k_m	Dissolution mass-transfer coefficient	m/s
$A_s(t)$	Exposed crystal surface area	m^2
C, C_{sat}	Salt mass concentrations in fluid and at saturation	$\mathrm{kg/m}^3$
ρ_w	Approximate density of pure water	$\mathrm{kg/m}^3$

Typical *illustrative* values used earlier: $k = 0.033\,\mathrm{W/(m\,K)}$, $A = 0.05\,\mathrm{m}^2$, $d = 0.02\,\mathrm{m}$, $T_{\mathrm{room}} = 25°\mathrm{C}$, $c_i = 2100$, $c_w = 4180\,\mathrm{J/(kg\,K)}$, $L_f = 333500\,\mathrm{J/kg}$, $M_{\mathrm{NaCl}} = 0.05844\,\mathrm{kg/mol}$, $\Delta H_{\mathrm{sol}} \approx 3900\,\mathrm{J/mol}$, $K_f = 1.86\,\mathrm{K\,kg/mol}$, $i_v \approx 2$ (*dilute limit*). Initial demonstration masses were $m_{i,0} = 0.200\,\mathrm{kg}$, $m_{w,0} = 0.100\,\mathrm{kg}$, $m_{s,0} = 0.050\,\mathrm{kg}$ and $T_0 = 0°\mathrm{C}$.

2 Step 1: energy in = energy out

The initial framing was

$$E_{\mathrm{in}} = E_{\mathrm{out}}, \tag{1}$$

$$E_{\mathrm{in}} - E_{\mathrm{out}} = \Delta U, \qquad U_{\mathrm{initial}} = U_{\mathrm{final}} \quad \text{if insulated and no work.} \tag{2}$$

For a completely insulated closed system, $E_{\mathrm{in}} = E_{\mathrm{out}} = 0$ and $\Delta U = 0$. For nonzero heat conduction and no work,

$$\frac{\mathrm{d}U}{\mathrm{d}t} = \dot{Q}_{\mathrm{in}}. \tag{3}$$

The equality $E_{\mathrm{in}} = E_{\mathrm{out}}$ only applies when accumulation is zero. For the transient system the correct general statement is Equation (2).

3 Step 2: heat entering through Styrofoam

In the simplified planar, quasi-steady one-dimensional conduction approximation,

$$\dot{Q}_{\text{Styrofoam}}(t) = \frac{kA}{d}(T_{\text{room}} - T(t)), \tag{4}$$

$$Q_{\text{in}}(t) = \int_0^t \frac{kA}{d}\left[T_{\text{room}} - T(\tau)\right] \mathrm{d}\tau, \tag{5}$$

$$H = \frac{kA}{d}, \tag{6}$$

$$\frac{\mathrm{d}U}{\mathrm{d}t} = H(T_{\text{room}} - T(t)). \tag{7}$$

With multiple wall sections indexed by j, each with thickness d_j and area A_j, use the more flexible extension

$$\dot{Q}_{\text{in}} = \sum_j \frac{k_j A_j}{d_j}(T_{\text{room}} - T). \tag{8}$$

The approximation neglects air-side film resistances, thermal bridges and Styrofoam transient heat storage. If relevant, replace H with a measured effective conductance or thermal resistance network.

4 Step 3: the ice and the latent heat

For the ice sensible heat (at fixed mass), and the latent heat consumed in melting,

$$\mathrm{d}Q_i = m_i c_i \,\mathrm{d}T, \tag{9}$$

$$\dot{Q}_{i,\text{sensible}} = m_i c_i \dot{T}, \tag{10}$$

$$\mathrm{d}Q_{\text{melt}} = L_f \,\mathrm{d}m_{\text{melt}}, \tag{11}$$

$$\dot{Q}_{\text{melt}} = L_f \dot{m}_{\text{melt}}, \tag{12}$$

$$\dot{m}_{\text{melt}} = -\dot{m}_i, \tag{13}$$

$$\dot{Q}_{\text{melt}} = -L_f \dot{m}_i. \tag{14}$$

The approximate *combined* energy balance (also including sensible heat of the water) used in the conversation is

$$H(T_{\text{room}} - T) = (m_i c_i + m_w c_w)\dot{T} - L_f \dot{m}_i. \tag{15}$$

Here ice melting has $\dot{m}_i < 0$, so the term $-L_f \dot{m}_i$ is positive.

5 Step 4: liquid water and mass conservation

The liquid-water sensible heating formula, at fixed water mass, is

$$Q_w = m_w c_w \Delta T, \tag{16}$$

$$\dot{Q}_{w,\text{sensible}} = m_w c_w \dot{T}. \tag{17}$$

If melting/freezing is the only water phase change, and no mass enters or exits,

$$\mathrm{d}m_w = -\mathrm{d}m_i, \tag{18}$$

$$\dot{m}_w = -\dot{m}_i, \tag{19}$$

$$m_w(t) + m_i(t) = m_{w,0} + m_{i,0} = M_w, \tag{20}$$

$$m_w(t) = m_{w,0} + m_{i,0} - m_i(t) = M_w - m_i(t). \tag{21}$$

Substituting this into Equation (15) gives

$$H(T_{\text{room}} - T) = [m_i c_i + (M_w - m_i)c_w]\,\dot{T} - L_f \dot{m}_i. \tag{22}$$

6 Step 5: enthalpy of dissolution of NaCl

For dissolved salt mass m_s, number of moles $n_s = m_s/M_{\text{NaCl}}$, and dissolution energy,

$$n_s = \frac{m_s}{M_{\text{NaCl}}}, \tag{23}$$

$$\dot{Q}_{\text{sol}} = \Delta H_{\text{sol}} \dot{n}_s = \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}} \dot{m}_s. \tag{24}$$

As NaCl dissolution in water is weakly endothermic, positive $\dot{m}_s$ absorbs energy. The approximate energy balance now reads

$$\boxed{H(T_{\text{room}} - T) = (m_i c_i + m_w c_w)\dot{T} - L_f \dot{m}_i + \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}} \dot{m}_s.} \tag{25}$$

Equivalently, eliminate m_w:

$$H(T_{\text{room}} - T) = [m_i c_i + (M_w - m_i)c_w]\,\dot{T} - L_f \dot{m}_i + \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}} \dot{m}_s. \tag{26}$$

Physical scope: this expression uses constant ice/liquid heat capacities, fixed reference enthalpy of fusion and an approximate constant heat of dissolution. For high precision at subzero temperatures use the phase-specific internal energy formulation in Section 11.

7 Step 6: freezing-point depression $i_v K_f b$

For a dilute solution,

$$\Delta T_f = i_v K_f b, \tag{27}$$

$$b(t) = \frac{m_s(t)/M_{\text{NaCl}}}{m_w(t)}, \tag{28}$$

$$T_f(t) = -i_v K_f b(t) = -\frac{i_v K_f m_s(t)}{M_{\text{NaCl}} m_w(t)}, \tag{29}$$

$$T(t) = T_f(t) \quad \textit{only when ice and brine coexist in equilibrium}, \tag{30}$$

and, substituting the mass balance,

$$\boxed{T = -\frac{i_v K_f m_s}{M_{\text{NaCl}}(M_w - m_i)}.} \tag{31}$$

For example, 0.005 kg dissolved NaCl in 0.100 kg liquid water gives

$$T_f = -2 \times 1.86 \times \frac{0.005/0.05844}{0.100} \approx -3.18^{\circ}\text{C}. \tag{32}$$

The dilute approximation is *not* quantitatively valid for concentrated NaCl brine. The NaCl–water eutectic lies near -21.1°C at about 23.3% NaCl by mass; there, measured phase-equilibrium data must replace $i_v K_f b$.

8 Step 7: salt dissolution rate and shrinking crystals

A mass-transfer-limited dissolution-rate law used in the conversation is

$$\dot{m}_s = k_m A_s (C_{\text{sat}} - C), \tag{33}$$

$$C(t) \approx \frac{\rho_w m_s(t)}{m_w(t)}, \tag{34}$$

$$\dot{m}_s \approx k_m A_s \left[C_{\text{sat}}(T) - \frac{\rho_w m_s}{m_w} \right]. \tag{35}$$

For a fixed number of similarly shaped shrinking crystals, assuming surface area scales as solid mass$^{2/3}$,

$$m_{\text{solid}}(t) = m_{s,0} - m_s(t) \quad \text{if no NaCl initially dissolved,} \tag{36}$$

$$A_s(t) = A_{s,0} \left(\frac{m_{s,0} - m_s(t)}{m_{s,0}} \right)^{2/3}, \tag{37}$$

so

$$\boxed{\dot{m}_s = k_m A_{s,0} \left(\frac{m_{s,0} - m_s}{m_{s,0}} \right)^{2/3} \left[C_{\text{sat}}(T) - \frac{\rho_w m_s}{m_w} \right].} \tag{38}$$

Domain of validity: $0 \le m_s < m_{s,0}$, undissolved salt must remain, and the brine must be undersaturated. A physically constrained evaporation-free, precipitation-free extension would use $\max(0, C_{\text{sat}} - C)$ and switch off dissolution at complete depletion. For general initial dissolved salt mass use $S_0 - m_s$ in place of $m_{s,0} - m_s$. For concentrated brines, use $C = m_s / V_{\text{brine}}$ with measured brine density rather than the approximation (34).

9 Step 8: fully unified *time-dependent* model

Define the shortcuts

$$W = M_w - m_i = m_w, \tag{39}$$

$$C_{\text{eff}} = m_i c_i + W c_w, \tag{40}$$

$$\alpha = \frac{i_v K_f}{M_{\text{NaCl}}}, \qquad \beta = \frac{\Delta H_{\text{sol}}}{M_{\text{NaCl}}}, \qquad H = \frac{kA}{d}. \tag{41}$$

Here α has units K kg/kg (in the indicated mass ratio) and β has units J/kg. The coupled balance and equilibrium relation are

$$H(T_{\text{room}} - T) = C_{\text{eff}} \dot{T} - L_f \dot{m}_i + \beta \dot{m}_s, \tag{42}$$

$$T = -\alpha \frac{m_s}{W}. \tag{43}$$

Differentiating (43) carefully, since $\dot{W} = -\dot{m}_i$,

$$\dot{T} = -\alpha \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{m_s}{W} \right) \tag{44}$$

$$= -\alpha \left(\frac{\dot{m}_s W - m_s \dot{W}}{W^2} \right) \tag{45}$$

$$= -\alpha \left(\frac{\dot{m}_s}{W} + \frac{m_s \dot{m}_i}{W^2} \right). \tag{46}$$

Substitute (46) into (42):

$$H(T_{\text{room}} - T) = -\alpha C_{\text{eff}} \left(\frac{\dot{m}_s}{W} + \frac{m_s \dot{m}_i}{W^2} \right) - L_f \dot{m}_i + \beta \dot{m}_s \tag{47}$$

$$= \left(\beta - \frac{\alpha C_{\text{eff}}}{W} \right) \dot{m}_s - \left(L_f + \frac{\alpha C_{\text{eff}} m_s}{W^2} \right) \dot{m}_i. \tag{48}$$

Solve for the ice-mass derivative:

$$\boxed{\dot{m}_i = \frac{\left(\beta - \frac{\alpha C_{\text{eff}}}{W} \right) \dot{m}_s - H(T_{\text{room}} - T)}{L_f + \frac{\alpha C_{\text{eff}} m_s}{W^2}}.} \tag{49}$$

For the exact previously proposed shrinking-crystal closure,

$$\boxed{\dot{m}_s = k_m A_{s,0} \left(\frac{m_{s,0} - m_s}{m_{s,0}} \right)^{2/3} \left[C_{\text{sat}}(T) - \frac{\rho_w m_s}{W} \right].} \tag{50}$$

The **full reduced time-dependent system** is therefore

$$\boxed{\begin{aligned} \dot{m}_s &= k_m A_{s,0} \left(\frac{m_{s,0} - m_s}{m_{s,0}} \right)^{2/3} \left[C_{\text{sat}}(T) - \frac{\rho_w m_s}{W} \right], \\ \dot{m}_i &= \frac{\left(\beta - \frac{\alpha C_{\text{eff}}}{W} \right) \dot{m}_s - H(T_{\text{room}} - T)}{L_f + \frac{\alpha C_{\text{eff}} m_s}{W^2}}, \\ T &= -\alpha \frac{m_s}{W}, \qquad W = M_w - m_i, \quad C_{\text{eff}} = m_i c_i + W c_w. \end{aligned}} \tag{51}$$

Specify $m_i(0)$ and $m_s(0)$ consistently with Equation (43). This *ice/brine equilibrium branch* ceases to apply when ice is exhausted, salt becomes saturated, or the mixture is significantly out of phase equilibrium.

9.1 Alternative single implicit equation (also derived earlier)

Eliminating T from the heat balance without solving explicitly for $\dot{m}_i$ gives

$$\boxed{H \left(T_{\text{room}} + \frac{\alpha m_s}{W} \right) = -\alpha C_{\text{eff}} \left(\frac{\dot{m}_s}{W} + \frac{m_s \dot{m}_i}{W^2} \right) - L_f \dot{m}_i + \beta \dot{m}_s.} \tag{52}$$

This implicit equation is algebraically equivalent to (49) paired with $T = -\alpha m_s / W$.

10 Step 9: *time-independent* equilibrium formulation

There are two distinct meanings of "time-independent": finite-time energy accounting between endpoints, and a true asymptotic steady state. They are not interchangeable.

10.1 Insulated equilibrium with $T_0 = 0°\text{C}$

For an insulated system, $Q_{\text{in}} = 0$, hence $U_f = U_0$. The approximate endpoint balance used previously, for initial ice and liquid water at the *reference temperature* $T_0 = 0°\text{C}$, is

$$\boxed{0 = (m_{i,f} c_i + m_{w,f} c_w) T_f - L_f (m_{i,f} - m_{i,0}) + \beta (m_{s,f} - m_{s,d,0}).} \tag{53}$$

With phase coexistence,

$$m_{w,f} = M_w - m_{i,f}, \tag{54}$$

$$T_f = -\alpha \frac{m_{s,f}}{M_w - m_{i,f}}. \tag{55}$$

If excess solid salt remains and the brine is saturated, impose

$$C_f = C_{\mathrm{sat}}(T_f), \tag{56}$$

whereas if all solid salt has dissolved, impose $m_{s,f} = S_0$; these are alternative regimes, *not* simultaneous constraints. Combine (53), (54), (55) and the appropriate salt condition to obtain the static state, provided it lies within the assumed phase regime.

10.2 Integrated heat leakage over a finite interval

Direct integration of the approximate balance (25) for constant c_i, c_w, L_f, β requires care: the term $C_{\mathrm{eff}}\dot{T}$ is *not* the derivative of $C_{\mathrm{eff}}T$ when masses change. We preserve the exact integral form of the approximate rate equation:

$$\boxed{\int_0^{t_f} H[T_{\mathrm{room}} - T(t)]\,\mathrm{d}t = \int_0^{t_f} C_{\mathrm{eff}}(t)\dot{T}(t)\,\mathrm{d}t - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}).} \tag{57}$$

The earlier endpoint formula was

$$\int_0^{t_f} H[T_{\mathrm{room}} - T(t)]\,\mathrm{d}t \overset{\text{earlier approximation}}{\approx} C_f T_f - C_0 T_0 - L_f(m_{i,f} - m_{i,0}) + \beta(m_{s,f} - m_{s,d,0}). \tag{58}$$

Correction: Equation (58) is not generally valid when C_{eff} varies. From integration by parts,

$$\int_0^{t_f} C_{\mathrm{eff}}\dot{T}\,\mathrm{d}t = [C_{\mathrm{eff}}T]_0^{t_f} - \int_0^{t_f} T\dot{C}_{\mathrm{eff}}\,\mathrm{d}t, \qquad \dot{C}_{\mathrm{eff}} = (c_i - c_w)\dot{m}_i. \tag{59}$$

Thus Equation (57), rather than the simplified endpoint replacement, must be used if following the original rate equation literally.

10.3 True steady state with heat leakage

At steady state $\dot{T} = \dot{m}_i = \dot{m}_s = 0$, so

$$H(T_{\mathrm{room}} - T_{\mathrm{eq}}) = 0, \tag{60}$$

$$\boxed{T_{\mathrm{eq}} = T_{\mathrm{room}}} \qquad (H > 0). \tag{61}$$

For $T_{\mathrm{room}} > 0°\mathrm{C}$, the long-time state has no ice in normal equilibrium (assuming enough time, no continuing cooling source and no other phase changes). Therefore the steady state *cannot* be obtained by enforcing an ice–brine freezing condition on a warm-room mixture.

11 A thermodynamically more consistent refinement

The previous model treats L_f as constant while separately using $c_i \neq c_w$. A thermodynamically consistent way to model phase change uses reference-specific energies (or enthalpies at approximately constant pressure). Choose $T_{\mathrm{ref}} = 0°\mathrm{C}$, set the specific enthalpy of ice at that reference to zero, and let liquid water have reference enthalpy L_f. In a first ideal-mixture approximation,

$$U_{\mathrm{approx}} = m_i c_i (T - T_{\mathrm{ref}}) + m_w[L_f + c_w(T - T_{\mathrm{ref}})] + \beta m_s. \tag{62}$$

Differentiate, using $\dot{m}_w = -\dot{m}_i$:

$$\dot{U}_{\text{approx}} = (m_i c_i + m_w c_w)\dot{T} + \left[(c_i - c_w)(T - T_{\text{ref}}) - L_f\right]\dot{m}_i + \beta \dot{m}_s, \tag{63}$$

$$H(T_{\text{room}} - T) = (m_i c_i + m_w c_w)\dot{T} - \left[L_f + (c_w - c_i)(T - T_{\text{ref}})\right]\dot{m}_i + \beta \dot{m}_s. \tag{64}$$

At $T = T_{\text{ref}}$ this agrees with the conversation's simpler balance, but at $T \neq T_{\text{ref}}$ it has a temperature-dependent fusion-energy contribution. The precise model should additionally incorporate salt-dependent brine enthalpy, solution density, mixing nonideality, NaCl liquidus data and hydration/solid phases where relevant.

11.1 Consistent integrated energy at arbitrary initial and final temperature

For the same simplified reference-energy state function,

$$\boxed{\int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = U_{\text{approx}}(t_f) - U_{\text{approx}}(0),} \tag{65}$$

where U_{approx} is given by Equation (62). Expanding it fully:

$$\begin{aligned}\int_0^{t_f} H(T_{\text{room}} - T)\,\mathrm{d}t = {} & m_{i,f} c_i (T_f - T_{\text{ref}}) - m_{i,0} c_i (T_0 - T_{\text{ref}}) \\ & + m_{w,f}[L_f + c_w(T_f - T_{\text{ref}})] - m_{w,0}[L_f + c_w(T_0 - T_{\text{ref}})] \\ & + \beta(m_{s,f} - m_{s,d,0}).\end{aligned} \tag{66}$$

This is the preferred endpoint equation for our *idealized, constant-coefficient* model. For a perfect insulation equilibrium, set the left-hand side to zero. For an exposed container over finite time the left side depends on the unknown temperature history and cannot generally be inferred from endpoints alone.

12 Model limits and practical next steps

1. **Initial temperature consistency.** The instantaneous-equilibrium condition $T = -\alpha m_s/m_w$ cannot accommodate pure ice+fresh water at a temperature above 0°C while ice is present. A non-equilibrium initial period, or piecewise regimes, may be required.

2. **Salt concentration.** The $i_v K_f b$ approximation is only suitable for relatively dilute brines. Use measured NaCl–H_2O phase boundary $T_f = f(w_s)$ for the intended concentration range.

3. **Dissolution kinetics.** Actual rates depend on stirring, crystal size, surface passivation and whether brine can circulate through the ice. k_m and $A_{s,0}$ should be measured or fitted.

4. **Container physics.** $H = kA/d$ assumes a planar, quasi-steady wall. Experimentally calibrating H is often more reliable than assuming nominal Styrofoam conductivity.

5. **Switching of phases.** Enforce $m_i \geq 0$, $0 \leq m_s \leq S_0$, and salt saturation. Ice/brine equilibrium, salt precipitation, fully melted liquid and all-solid or eutectic regimes each need appropriate closure equations.

6. **Validation.** Compare predicted temperature-time curves and equilibrium endpoints with experiments at several salt-to-water ratios, starting masses, initial temperatures and insulation thicknesses.

13 Quick index of principal equations

Conduction: (4); ice/water conservation: (21); latent energy: (15); NaCl energy: (24) and (25); freezing depression: (29)–(31); dissolution kinetics: (33)–(38); dynamic derivatives: (46) and (51); insulated equilibrium: (53)–(56); finite-time energy: (57); true room-coupled steady state: Section 9.3; thermodynamically consistent update: (64)–(66).